higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

ENGINEERING PRACTICE AND MAINTENANCE
NQF LEVEL 3

(6021003)

20 November 2019 (X-Paper)
09:00–12:00

This question paper consists of 8 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Read ALL the questions carefully.
2. Answer ALL the questions.
3. Number the answers according to the numbering system used in this question paper.
4. Start each question on a NEW page.
5. Write neatly and legibly.

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QUESTION 1

1.1 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.1.1–1.1.5) in the ANSWER BOOK.

1.1.1 Splash lubrication is done by hand.

1.1.2 One needs to modify existing design solutions in order to improve a product, process or system.

1.1.3 Simulations can be used to diagnose faults and their causes before the equipment or machinery is in operation.

1.1.4 A symptom is a warning sign that tells us that there is a fault in a system.

1.1.5 A plant is not a factory or building in which an industrial process happens such as manufacturing something.

(5 × 1) (5)

1.2 Choose a description from COLUMN B that matches a term in COLUMN A. Write only the letter (A–G) next to the question number (1.2.1–1.2.5) in the ANSWER BOOK.

COLUMN A		COLUMN B	
1.2.1	Waste disposal	A	product, parts or system has broken
1.2.2	Safe working conditions	B	oil pump is situated in the reservoir
1.2.3	Splash lubrication system	C	look and listen for failures
1.2.4	Random search method	D	helps to find the design solution
1.2.5	Fault	E	label all containers with approved labels
		F	unplanned, haphazard and aimless
		G	heavy objects are confined to lower shelves

(5 × 1) (5)

1.3 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.3.1–1.3.5) in the ANSWER BOOK.

1.3.1 Developing … helps solve the problem.

A defects
B ideas
C images
D work

1.3.2 Written or printed information sources that provide a variety of information:

A A patent
B Volts
C Measuring
D Consumables

1.3.3 The principle of fault-finding diagnostics is a process used to investigate symptoms of faults.

A Reliability
B Software
C Safety first
D Additives

1.3.4 … will inform you how often the machine has been maintained.

A Warn brakes
B Service records
C Fuel records
D Lack of a gear box

1.3.5 A manual feed lubrication system uses … to lubricate.

A a dry sump lubrication system
B an oil can
C force feed
D a splash lubrication system

(5 × 1) (5)

1.4 Give ONE word or a term for each of the following descriptions by choosing a word or term from the list below. Write only the word or term next to the question number (1.4.1–1.4.5) in the ANSWER BOOK.

force; include; lubricant; draw; diagnose; problem; failure

1.4.1 Safe working practices … storing equipment and supplies safely.

1.4.2 Any substance that is used to provide coating or film between moving metal parts.

1.4.3 An ohmmeter may be used to … fault in the earthing or bonding of exposed conductive parts in an electric circuit.

1.4.4 The product or system does not perform as expected.

1.4.5 One of the skills needed by engineers is to develop ideas to help solve a …

(5 × 1) (5)
[20]

QUESTION 2

2.1 Testing and evaluating our solutions is part of abstraction and synthesis.

List FOUR methods of testing and evaluation. (4)

2.2 Distinguish between the following:

2.2.1 Design for quality and design for export

2.2.2 Bench marking and best practices

(2 × 2) (4)

2.3 Engineers from different companies in engineering have common skills that they need to be able to perform.

List FOUR skills of engineers from the working environment. (4)

2.4 Write down FOUR advantages of using CAD (computer-aided design) in engineering. (4)

2.5 Describe each of the following:

2.5.1 Manuals in engineering

2.5.2 Workplace specifications

(2 × 2) (4)
[20]

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QUESTION 3

3.1 Name THREE methods to identify symptoms of faults. (3)

3.2 Collect data and information is very important to keep records, and to help you diagnose or judge the cause of the fault.

List FOUR types of data and information that can help you identify the symptoms of a fault. (4)

3.3 Techniques are used in a systematic way to find faults and find out what is wrong.

Write down the name of a technique that is used or that may be used to assist in identifying the problems described in the following sentences:

3.3.1 The bulbs in my car lights are always blowing.

3.3.2 My car has started to bounce more than it used to when driving over bumps in the road.

3.3.3 I have noticed white powder around the car battery.

3.3.4 The owner's manual of my car suggests that when I hear a knocking sound coming from the wheels when I turn, is a sign that the CV joints are worn and need replacing.

3.3.5 Occasionally, when I have been waiting in traffic for a while, a cloud of blue smoke is expelled from my exhaust.

(5 × 1) (5)

3.4 Differentiate between the following:

3.4.1 Simulation and setting up the system for failure (4)

3.4.2 Natural and synthetic lubricants (2)

3.5 What are the positive aspects of a manufacturer's service diagnostic data? (2)

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QUESTION 4

4.1 Explain each of the following terms used in engineering:

4.1.1 Extraction plants

4.1.2 Force feed lubrication system

4.1.3 Power plant

4.1.4 Channels of notification

4.1.5 Analysis of solutions

(5 × 2) (10)

4.2 A car owner visits you in your workplace and asks you to repair his faulty engine.

State FIVE steps you will follow in the five-step approach to fault-finding in order to solve the problem. (5)

4.3 Copy the table below in your ANSWER BOOK and show the difference between systematic and random search methods by completing the table.

Write down TWO characteristics/differences of each method.

SYSTEMATIC SEARCH METHOD		RANDOM SEARCH METHOD
e.g.	Planned, logical and has an aim	Unplanned, haphazard and aimless
4.3.1		
4.3.2		

(2 × 2) (4)

4.4 Give ONE example of a technical aid that is used as one of common aids to fault location. (1)

[20]

QUESTION 5

Read the passage below and answer the questions.

EQUIPMENT FAILURES AND THEIR CONTRIBUTION TO INDUSTRIAL INCIDENTS AND ACCIDENTS IN THE MANUFACTURING INDUSTRY

Accidental events in manufacturing industries can be caused by many factors, including work methods, lack of training, equipment design, maintenance and reliability. This study is aimed at determining the contribution of failures of commonly used industrial equipment, such as machines, tools and material handling equipment, to the chain of causality of industrial accidents and incidents.

Based on a case study which aimed at the analysis of an existing pulp and paper company's accident database, this paper examines the number, type and gravity of the failures involved in these events and their causes. Results from this study show that equipment failures had a major effect on the number and severity of accidents accounted for in the database: 272 out of 773 accidental events were related to equipment failure, where 13 of them had direct human consequences. Failures that contributed directly or indirectly to these events were analysed, such as accidents; availability; maintenance policy; manufacturing, pulp and paper company; reliability; safety; and a safety and failure database.

[Source: *www.ncbi.nim.nih.gov/pubmed/26652772*]

5.1 Name THREE factors that cause accidental events in manufacturing industries. (3)

5.2 What are the contributing failures in industrial equipment? (3)

5.3 Explain the objectives of this case study. (2)

5.4 The manufacturer's specifications, company policy and legislation prescribe safety measures when operating hazardous equipment.

What are the prescribed steps of safety measures when operating hazardous equipment? (4)

5.5 State FOUR roles of lubricants in machine components. (4)

5.6 Differentiate between a *hydrostatic principle* and a *hydrodynamic principle*. (4)

[20]

TOTAL: 100

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